MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GASEOUS DISCHARGE

TYPE 7617

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Relaxation oscillator tube

See figure 1

Mounting position: Any

Weight: 0.353 ounce (10 grams) nominal

ABSOLUTE RATINGS:

Parameter:	TA	Barometric pressure, reduced
Unit:	°C	mmHg
Maximum:	+55	87
Minimum:	-20	---
TEST CONDITIONS:	---	---

GENERAL:

Qualification - Required

(F) Holding period - See quality conformance inspection, part 1

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
---	Pulse amplitude	6	T = +25°C	0.65	II	eo	45	55	v
---	Pulse recurrence rate	6	T = +25°C	0.65	II	prr	1,785	2,415	---
(F) ---	Holding period	---		---	---	t	22	---	days
(F) ---	Holding period end points:	3							
---	Pulse recurrence rate	---		0.65	II	Δprr	---	3	%
---	Pulse amplitude	---		0.65	II	Δeo	---	3	%
	Quality conformance inspection, part 2								
3305	Leakage current	---	Eb = 200 Vdc; T = 25° +5°C; relative humidity = 90 +5%	---	---	LIb	---	0.2	uAdc
	Quality conformance inspection, part 3								
(F) ---	Life test	1	Group D	---	---	t	1,000	---	hrs
---	Life-test end points:								
---	Pulse recurrence rate	---		---	---	Δprr	---	4	%
---	Pulse amplitude	---		---	---	Δeo	---	6	%
1031	Variable-frequency vibration	7	No voltages applied	---	---	---	---	---	---
1041	Shock	2,7	Hammer angle = 20°	---	---	---	---	---	---
(F) 1026	Temperature range, operating:	6,7							
---	Room temperature	---	T = +25°C	---	---	prr eo	1,785 45	2,415 55	--- v
	Cold	4,5	T = -20°C	---	---	---	---	---	---
	Hot	4,5	T = +55°C	---	---	---	---	---	---

NOTES:

1. The circuit as shown on figure 2 shall be used for life-test operation.
2. Electrical samples shall not be used for this test. Ten additional samples shall be submitted for shock, specified pulse test only. Criterion for passing this test shall be compliance after test of at least 80 percent of the tubes with the initial requirements for pulse recurrence rate (prr) and pulse amplitude (eo).
3. (F) All tubes shall pass these tests. Testing shall be performed 22 or more days after initial production tests are made on the same tubes. A tube that fails these tests may be retested after an additional 22 or more days with the provision that the values of Δeo and Δprr do not deviate by more than 3 percent from the values at the beginning of the last 22-day period, and that the initial limits for these tubes are still met.
4. Variation in pulse amplitude (eo) over the temperature range -20° to +55°C shall not exceed ±6 percent of the values of pulse height at 25°C.
5. Changes in prr for temperatures between +55° and -20°C shall be -0.05 percent ±0.05 percent per °C.
6. Test shall be made with circuit as shown on figure 3.
7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 99
DLA - ES

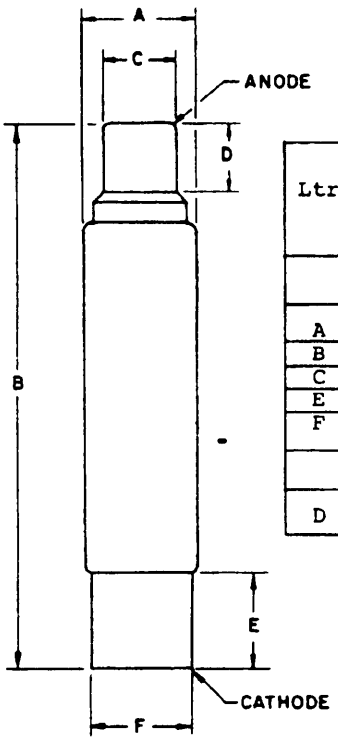
User activities:
Army - ME, AR, AV
Navy - AS, OS, MC

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3252)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, Part 2		
A		.500 (12.70)
B	2.438 (61.93)	2.562 (65.07)
C	.245 (6.22)	.255 (6.48)
E	.594 (15.09)	.656 (16.66)
F	.468 (11.89)	.488 (12.40)
Reference dimension.		
D	.250 (6.35)	

NOTE:
Mechanical configuration and materials used may vary, provided all specified dimensions are met.

FIGURE 1. Outline drawing of electron tube type 7617.

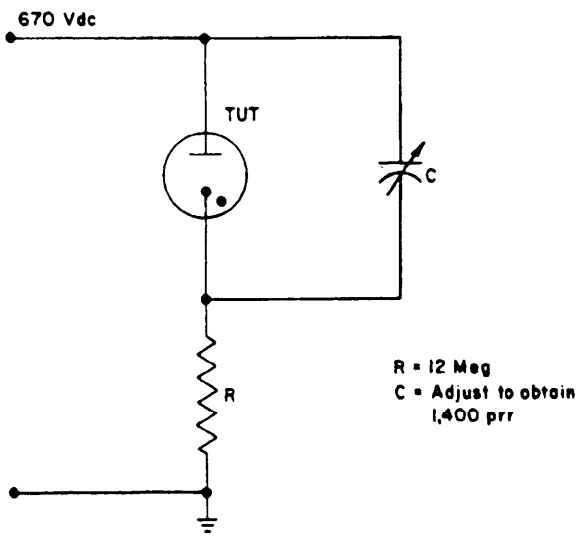
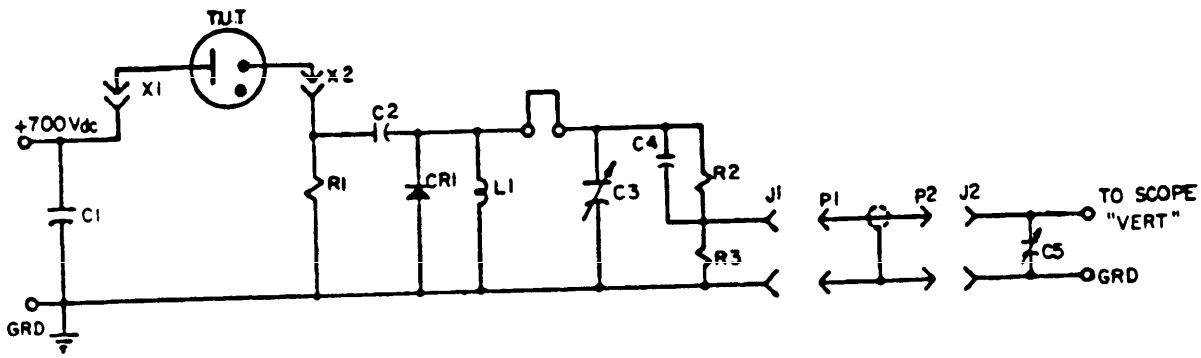


FIGURE 2. Circuit for life-test operation.

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C1 - 0.01 μF 1,000 Vdc
C2 - 160 pF ± 1% 500 Vdc
C3 - 2.5 to 13 pF
C4 - 33 pF ± 1% 500 Vdc
C5 - 7 to 45 pF
CR1 - 1N538
J1 and J2 - UG-912/U
L1 - 2 mH ± 1% at 500 kHz
P1 and P2 - UG-260/U BNC on 4 ft. RG-62/U coax cable
R1 - 3.5 MEG ± 0.05% TC ± 0.002%/°C.
R2 - 70kΩ ± 0.05% TC ± 0.002%/°C.
R3 - 25kΩ ± 0.05% TC ± 0.002%/°C.

FIGURE 3. Circuit used for measuring pulse rate frequency and pulse amplitude.

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